Outdoor Thermal Comfort and Recreational Walking Thresholds Among Older Adults in Subsidized Housing: Insights from Strava

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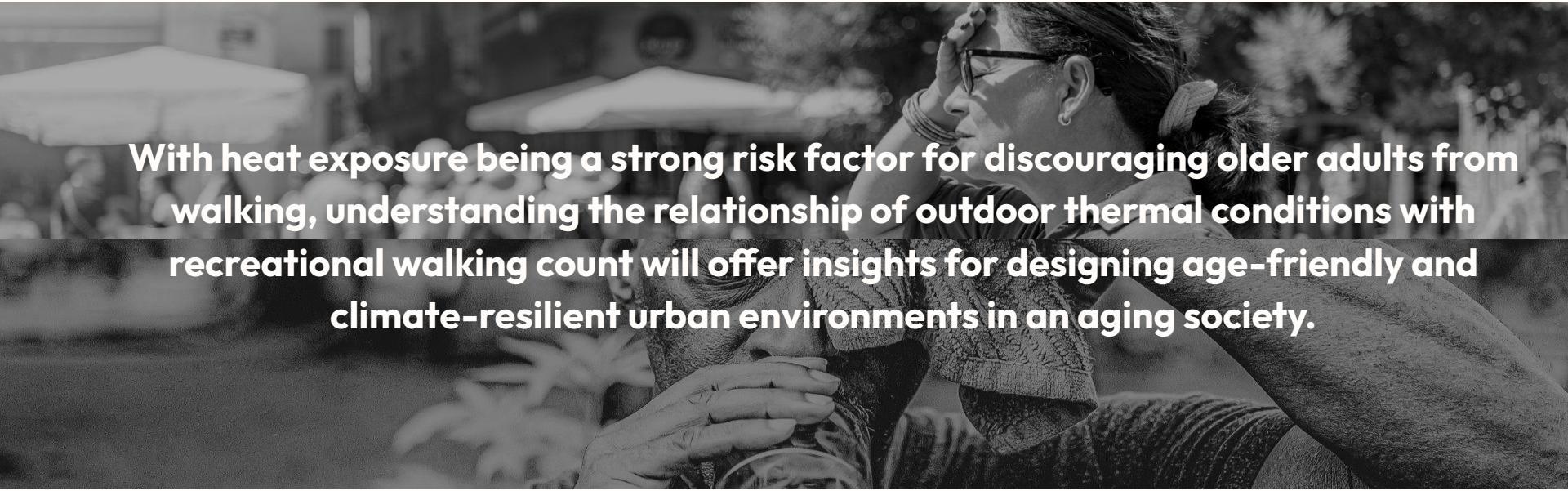
01 Introduction

- **Extreme heat poses increasing risks for outdoor physical activity**
  - Walking is important for healthy aging
  - Older adults are particularly vulnerable to heat
  - Physical activity is strongly influenced by thermal conditions

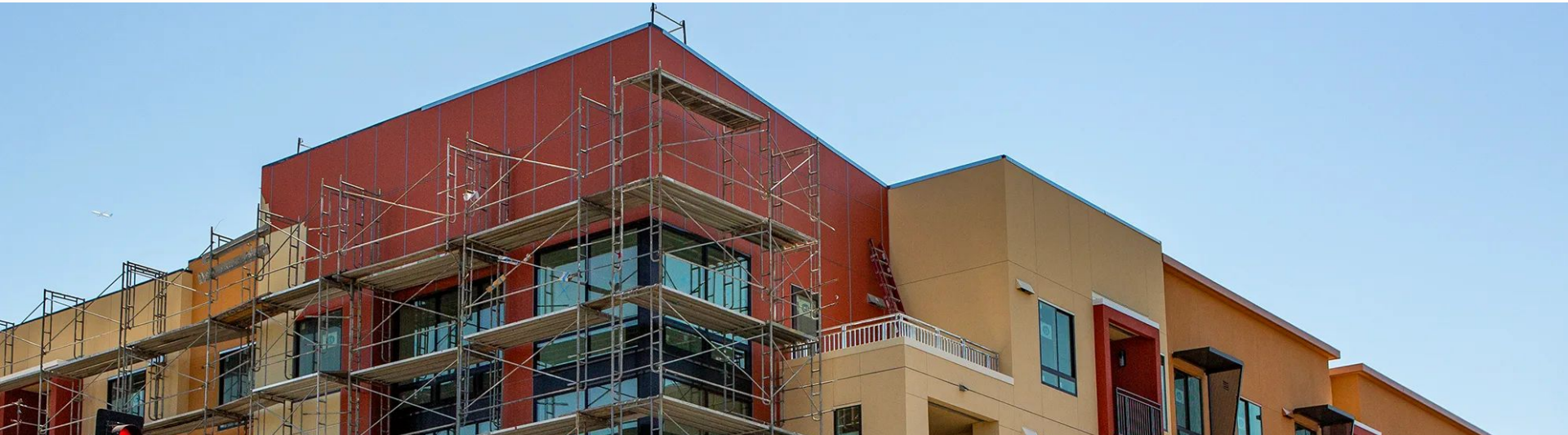
01 Introduction



With heat exposure being a strong risk factor for discouraging older adults from walking, understanding the relationship of outdoor thermal conditions with recreational walking count will offer insights for designing age-friendly and climate-resilient urban environments in an aging society.

01 Introduction

- **Environmental contexts of Subsidized housing**
  - Subsidized housing provides affordable rental housing for low-income households, older adults, and vulnerable populations
  - These communities often face environmental disadvantages (Chakraborty et al., 2022).
  - Residents may rely more on walking and outdoor activities for daily mobility and recreation, making thermal conditions particularly important



01 Introduction

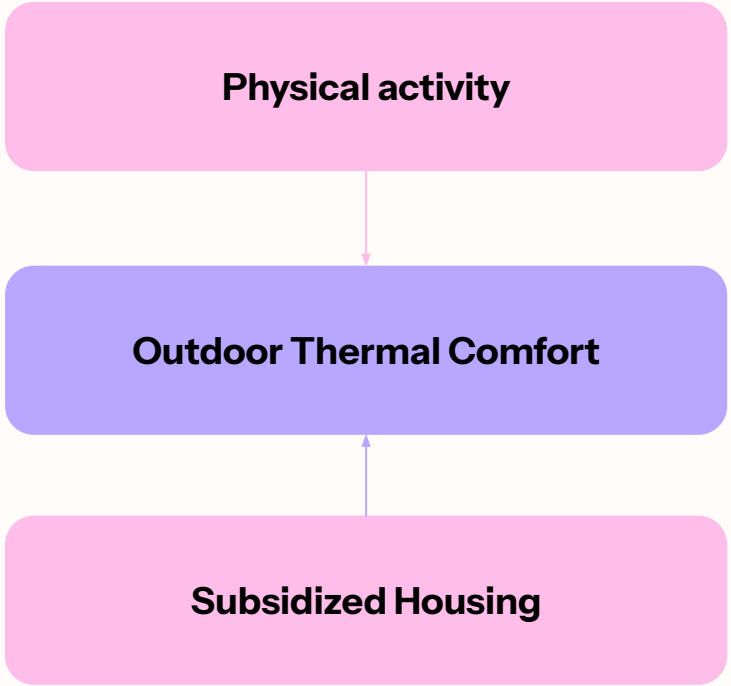
Understanding thermal conditions around subsidized housing is critical for assessing heat exposure risks and walking behavior in vulnerable communities.

01 Introduction - Research Gap

- **Existing studies have shown:**
  - Non-linear relationships between temperature and physical activity
    (e.g., activity increases with temperature until a threshold and declines beyond it) (Lanza et al., 2022)
  - Residents in subsidized housing often experience challenges related to neighborhood quality and environmental conditions
- **However :**

  Thermal exposure measurement
  - Few examine integrated thermal comfort indicators *(e.g., energy balance or heat stress models)*

  Housing environment inequality
  - Limited evidence on walking behavior and heat exposure around subsidized housing environments



01 Introduction - Study Objectives

- **This study aims to:**

1. **Identify thermal comfort thresholds for old adults**
   - Examine how outdoor thermal comfort relates to recreational walking counts.

2. **Compare housing environments**
   - Examine differences in recreational walking /and outdoor thermal comfort patterns around **subsidized housing vs apartment housing**

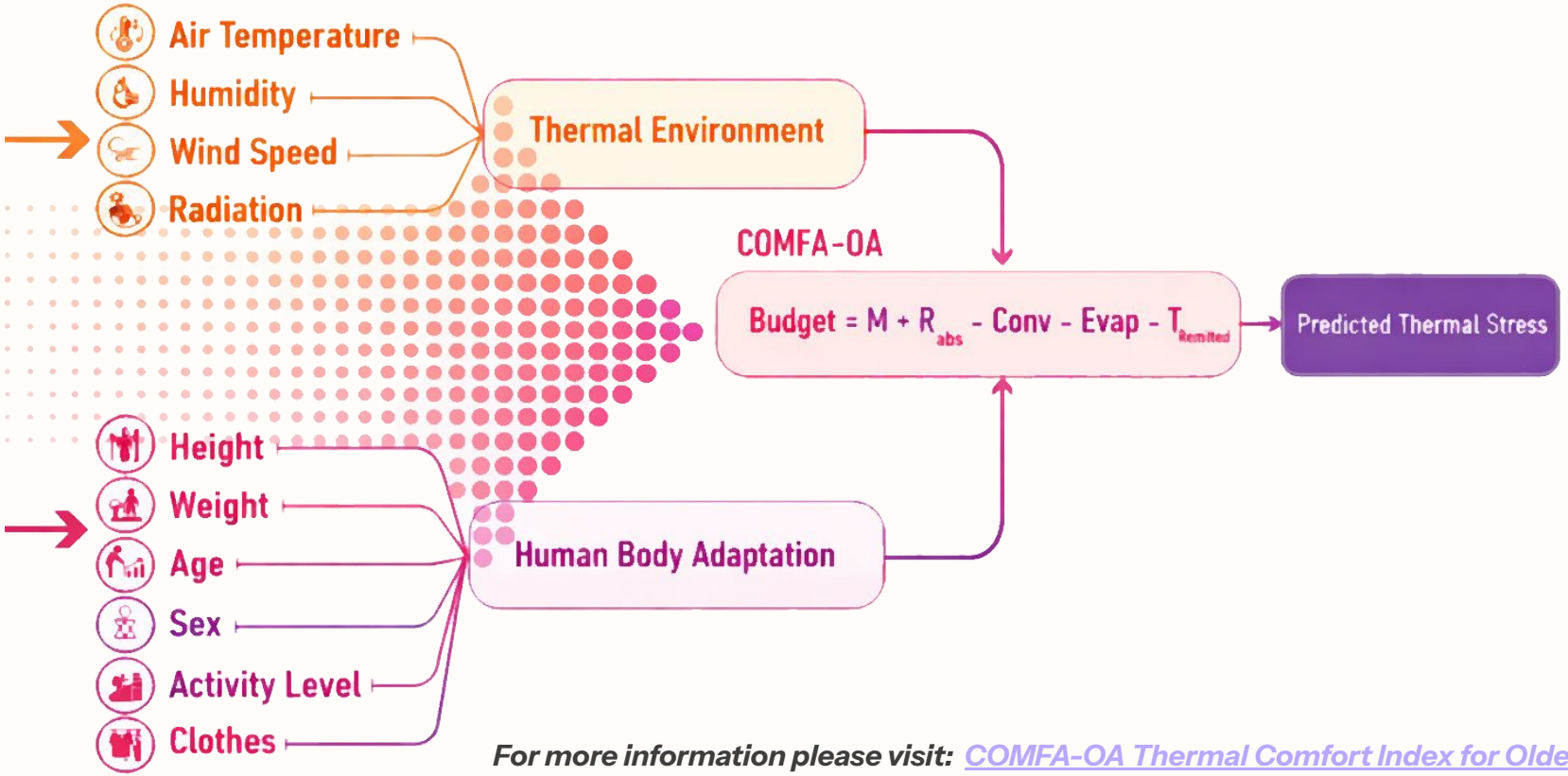
02 Method

- **Step1:** Identify the non-linear relationship between outdoor thermal comfort and older adult recreational walking.
  → Using a novel energy balance model ***COMFA-OA***
  → Zero-inflated negative binomial ***(ZINB)*** models

- **Step2:** Construct comparable housing samples.
  → Use Propensity ***Score Matching (PSM) to match apartment housing areas***
  → Match on ***demographic & SES variables***

- **Step3:** Derive thermal comfort thresholds.
  → Apply these thresholds to compare walking patterns around:
  - ***Subsidized housing areas***
  - ***Matched apartment housing areas.*** Within walkable buffers

02 Method- What is COMFA-OA



For more information please visit: [COMFA-OA Thermal Comfort Index for Older Adults]

02 Method- Data and Key Variables

Variable	Description	Unit	Data Source	Spatial Resolution
Recreational walking count	Strava Metro recreational walking activity aggregated per street segment	count	Strava Metro	street segment
COMFA-OA thermal comfort level	Outdoor thermal stress (W/m²)	W/m2	UMEP simulation	30m
Population aged 55+	Proportion of residents aged 55 years or older	%	ACS 2023 (5-Year)	
Land-use mix	Shannon entropy of NLCD land-cover composition within 400 m buffers around street segments	index	NLCD 2023 land cover raster	30 m raster; summarized within 400 m buffers
Access to destinations				
Access to health & aged care	Density of health and aged-care POIs	POI/km²	OpenStreetMap	within 400m buffer
Access to daily services	Density of retail and daily service POIs	POI/km²	OpenStreetMap	within 400m buffer
Access to social facilities	Density of social facility POIs	POI/km²	OpenStreetMap	within 400m buffer
Access to public services	Density of public service POIs	POI/km²	OpenStreetMap	within 400m buffer
Access to public transport	Density of public transport stop POIs	POI/km²	OpenStreetMap	within 400m buffer
Street connectivity				
Intersection density	Density of intersections within 400 m buffers	intersections/km	OpenStreetMap	within 400m buffer
Dead-end segment	Binary indicator of cul-de-sac street segment	0/1	OpenStreetMap	segment attribute
Street type				
Arterial	Arterial road classification	categorical	OpenStreetMap	street segment
Highway	Highway road classification	categorical		
Local	Local street classification	categorical		
Pedestrian	Pedestrian street classification	categorical		
Other	Other road types	categorical		

02 Method- Statistical analysis

The Zero-inflated negative binomial (ZINB) model consists of two response modeling components:

1) a negative binomial count model describing nonzero recreational walking counts and

$$\log(\mu_i) = \log(len_i) + f_8(COMFA_i) + \boldsymbol{x}_i^\top \boldsymbol{\beta} + \boldsymbol{s}_i^\top \boldsymbol{\gamma}$$

2) a binary logistic regression model describing probability of no recorded walking activity accounting for the zero-inflation.

$$\text{logit}(\pi_i) = g_4(COMFA_i) + \boldsymbol{w}_i^\top \boldsymbol{\alpha} + \boldsymbol{s}_i^\top \boldsymbol{\delta}$$

Based on the fitted relationship, we applied piecewise regression to identify thermal stress thresholds where walking activity changes sharply.

03 Results

3.1 Descriptive statistics

Figure 1 illustrates the spatial distribution of outdoor thermal comfort for older adults in Harris County during the period of July to September 2023.

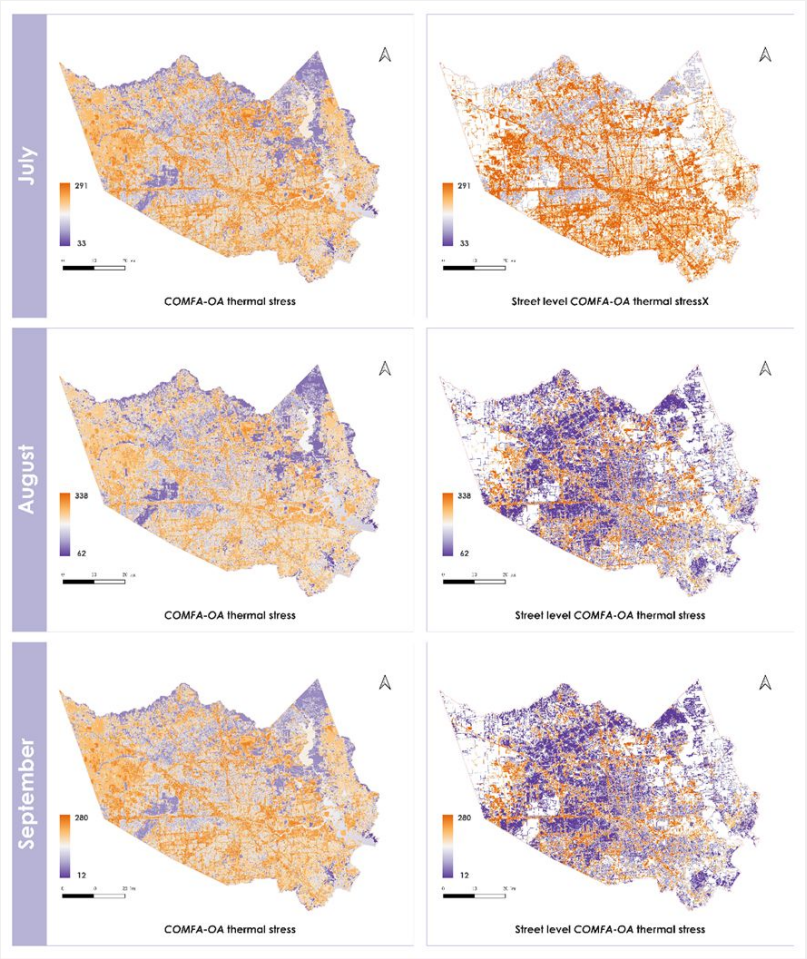


Figure 1 COMA-OA thermal stress for older adults in Harris County, July to September 2023.

03 Results

3.2 Identification of thermal stress thresholds related to recreational walking count

- Below Breakpoint 1 → Gradual decline
- Between B1 and B2 → Moderate sensitivity
- Above Breakpoint 2 → Sharp decline

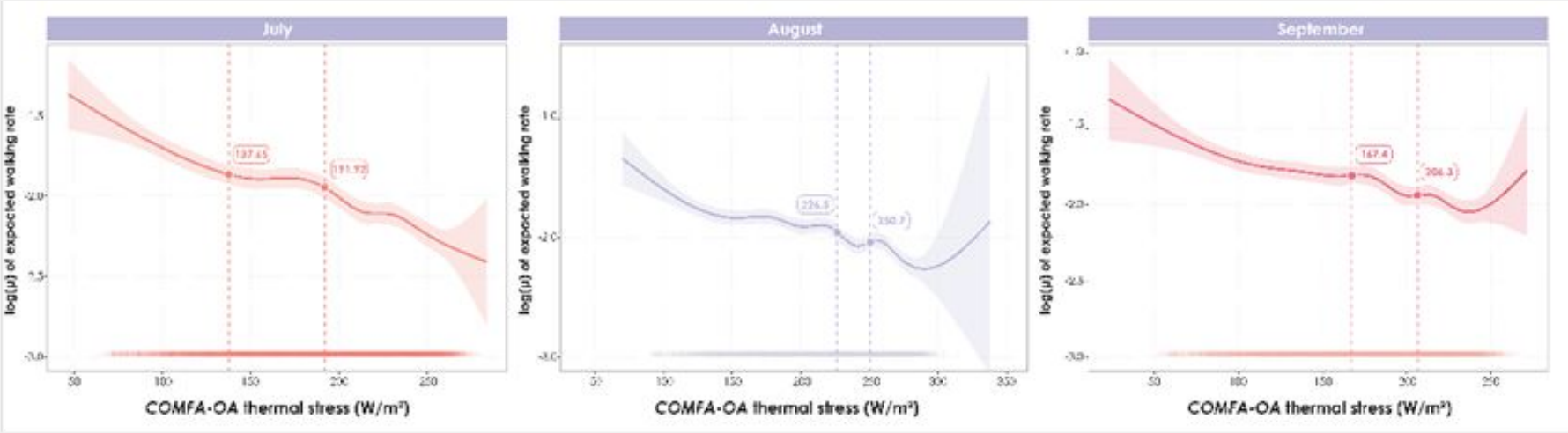


Figure 2 COMA-OA thermal stress for older adults in Harris County, July to September 2023.

03 Results

3.3 Differences in outdoor thermal stress and recreational walking activity across subsidized and apartment housing

PSM results:

Apartments were matched with nearest neighbor matching based on sociodemographic characteristics in a 1:1 ratio, including **share of low-income population, non-white population, and population aged over 55 years old.**

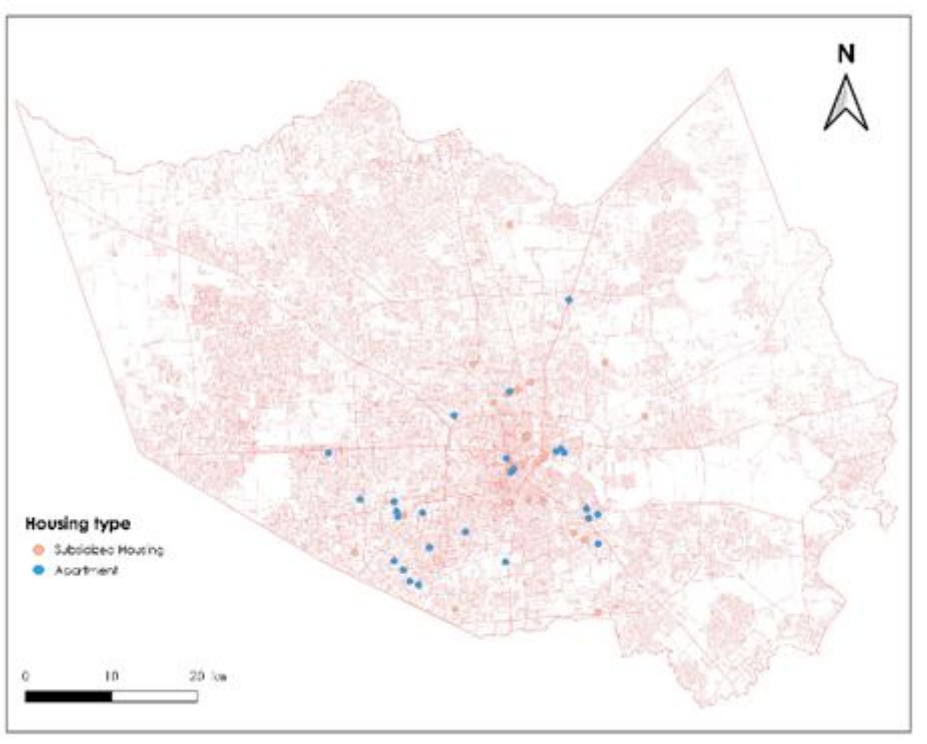


Figure 3 Distribution of post-PSM results in Harris County

03 Results

3.3 Differences in outdoor thermal stress and recreational walking activity across subsidized and apartment housing

COMFA-OA value comparison results:

Across all three months, no statistically significant differences were evident, with street segments around subsidized and apartment housing showing similar distributions of outdoor thermal stress.

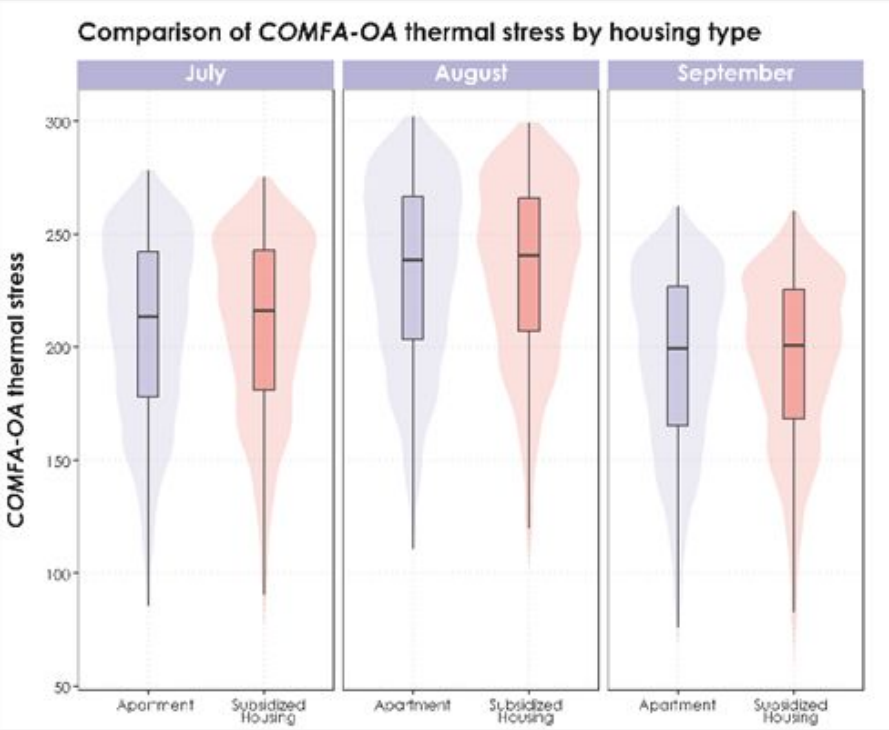


Figure 4 COMFA-OA thermal stress by housing type

03 Results

3.3 Differences in outdoor thermal stress and recreational walking activity across subsidized and apartment housing

Walking intensity comparison results:

In general, walking intensity adjacent to subsidized housing was higher than that near apartment housing within the lower and intermediate energy budget regimes, with small-to-moderate effect sizes (Cliff's d from 0.20 to 0.70).

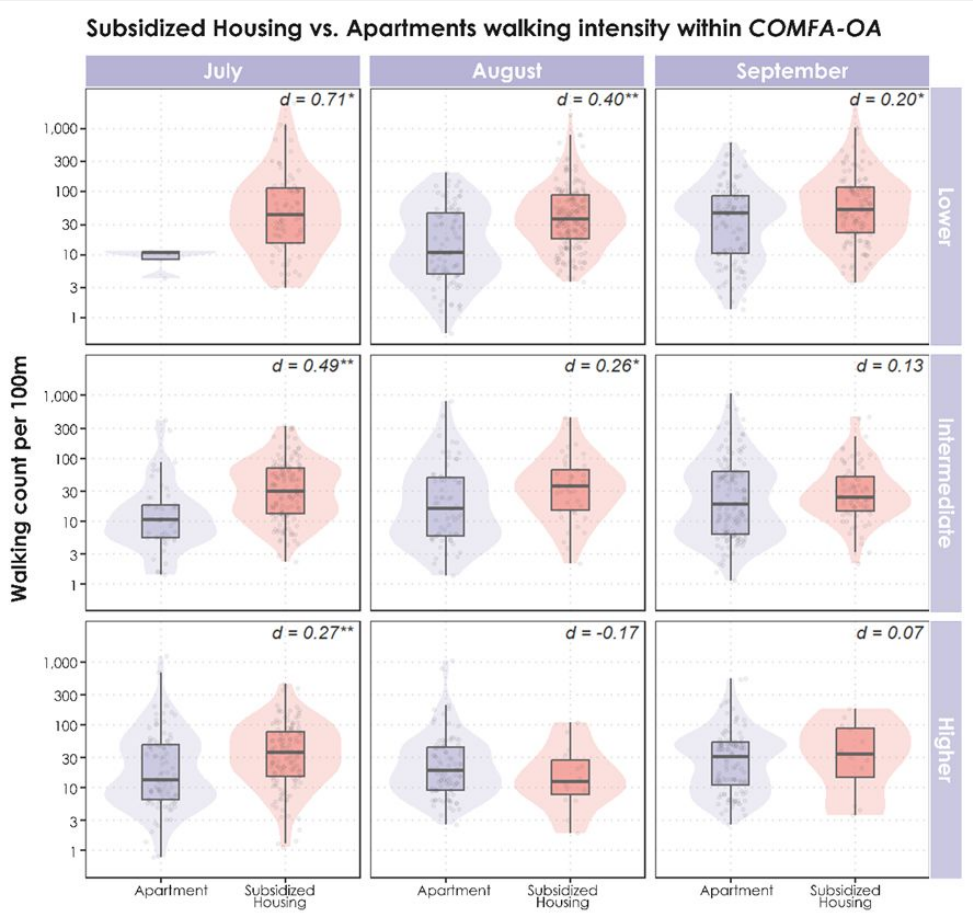


Figure 5 Comparison of walking counts according to COMFA-OA-based energy budget regime and housing type

04 Discussion– Identify thermal comfort thresholds for old adults

*According to the original COMFA framework, energy budget values between **201 and 340 W/m²** are classified as the "dangerous" heat exposure zone.*

Such misalignment suggests a lag between heat risk thresholds and behavioral adjustment.

→ **a). Limited awareness or delayed perception of heat-related risk during outdoor activity.**

→ **b). Acclimatization, repeated exposure enables individuals to tolerate higher levels of thermal stress before modifying behavior.**

What the Budget Values Mean for the Heat Health of an Adult (W/m^2)	
65 - 120	Caution
121 - 200	Extreme Caution
201- 340	Dangerous
> 341	Extremely Dangerous

For more information please visit: Dr. Robert Brown's Website

COMFA — Design with Microclimate

04 Discussion – Compare housing environments

01

Thermal exposure was similar across housing types

No significant differences in outdoor thermal stress between subsidized and apartment housing environments.

02

Walking activity was higher near subsidized housing

Observed under lower and intermediate thermal stress regimes. Possible explanation: greater reliance on walking and public transit

03

Heat strongly discourages recreational walking

When thermal stress reached higher energy budget regimes, walking activity declined sharply across both housing types

04 Discussion - Limitation

01 Data representation

Strava data may underrepresented older adults' walking activity.

02 Temporal and spatial aggregation

Walking counts and thermal stress were measured at different temporal and spatial scales.

03 Residential attribution

Street-level walking activity may not fully reflect the behavior of nearby residents.

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Thank you!